

1.9.3

Bhoomika L - EE25BTECH11014

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Question

$AOBC$ is a rectangle whose three vertices are $(0, -3)$ $(0, 0)$ $(4, 0)$. The length of its diagonal is_____

Theoretical Solution

Given the points **A**, **O** and **B** :

Point	vector
Point A	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$
Point O	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$
Point B	$\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

Table: Position Vectors of the points on rectangle.

Theoretical Solution

Determining the Coordinates of Point C:

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} \quad (1)$$

Since **C** is opposite to **O** in the rectangle,

$$\mathbf{C} = \mathbf{A} + \mathbf{B} \quad (2)$$

$$\Rightarrow \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (3)$$

$$\therefore \mathbf{C} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (4)$$

Theoretical solution

We know that the length of the diagonal vector is magnitude of the vector $\mathbf{C}$.

Let $\mathbf{P} = \mathbf{C}$

$$\mathbf{P} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (5)$$

$$|\mathbf{C}| = \sqrt{\mathbf{C}^T \cdot \mathbf{C}} \quad (6)$$

Theoretical Solution

$$\mathbf{c}^T \cdot \mathbf{c} = \begin{pmatrix} 4 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = 4^2 + (-3)^2 = 16 + 9 = 25 \quad (7)$$

$$|\mathbf{c}| = \sqrt{25} = 5 \quad (8)$$

Therefore the length of the diagonal is 5.

```
#include <stdio.h>

// Function to calculate point C of rectangle
// A=(Ax,Ay), B=(Bx,By), D=(Dx,Dy)
// Returns Cx and Cy via pointers
void findC(float Ax, float Ay, float Bx, float By, float Dx,
          float Dy, float *Cx, float *Cy) {
    *Cx = Bx + Dx - Ax;
    *Cy = By + Dy - Ay;
}
```



```
import matplotlib.pyplot as plt

# Define rectangle vertices
A = (0, 0)
B = (4, 0)
C = (4, -3)
O = (0, -3)

# Rectangle points in order for plotting closed shape
rectangle_x = [A[0], B[0], C[0], O[0], A[0]]
rectangle_y = [A[1], B[1], C[1], O[1], A[1]]
```

```
# Plot rectangle
plt.plot(rectangle_x, rectangle_y, 'g--', label=Rectangle)

# Plot diagonal AC
plt.plot([A[0], C[0]], [A[1], C[1]], 'b-', label=Diagonal)

# Mark all points
points = [A, B, C, O]
for point in points:
    plt.scatter(point[0], point[1], color='red')
    plt.text(point[0], point[1], f'{point}', fontsize=10, ha='
        left', va='bottom')
```

```
# Axis labels and grid
plt.xlabel(X-axis)
plt.ylabel(Y-axis)
plt.title(Rectangle AOBC with Diagonal)
plt.legend()
plt.grid(True)
plt.axis(equal)

# Save and show
plt.savefig(..Figs/fig2.png)
plt.show()
```

Plot

